

Cumulative Paper 1 — mark scheme

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Marks in brackets are awarded line by line: a line of correct method scores even when the arithmetic after it goes wrong. Follow through an earlier wrong answer wherever the later method is sound. The note under each question names the specific mistake that question was built to catch — that is what this booklet is for.

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- 1** $504 = 2 \times 252 = 2 \times 2 \times 126 = 2 \times 2 \times 2 \times 63 = 2 \times 2 \times 2 \times 7 \times 9$ **[1]** [2]
 $= 2^3 \times 3^2 \times 7$ **[1]**
A complete tree with the answer left as $2 \times 2 \times 2 \times 3 \times 3 \times 7$ scores 1 of 2 — the question asked for index notation.
- 2** $84 = 2^2 \times 3 \times 7$ $126 = 2 \times 3^2 \times 7$ [2]
 (a) HCF = $2 \times 3 \times 7 = 42$ **[1]**
 (b) LCM = $2^2 \times 3^2 \times 7 = 252$ **[1]**
The classic swap is 252 for the HCF and 42 for the LCM. An HCF can never be larger than the smaller number — say that out loud when marking it.
- 3** (a) $-60 \div (-4) = 15$ **[1]** [2]
 (b) $(-8) \times 9 = -72$ **[1]**
(b) catches the student who makes $(-2)^3$ positive because “two negatives make a positive”. An odd power keeps the sign; an even power loses it.
- 4** $\sqrt[3]{-216} = -6$ and $\sqrt{196} = 14$ **[1]** [2]
 $-6 + 14 = 8$ **[1]**
An answer of 20 means both roots were taken positive. A cube root of a negative number is negative — that is the whole distinction being tested here.
- 5** (a) $3^9 \div 3^6 = 3^3$ **[1]** [2]
 (b) $4^6 \times 4^1 \div 4^5 = 4^7 \div 4^5 = 4^2$ **[1]**
Watch for $(4^3)^2$ being read as 4^5 — a power of a power multiplies. Bare 27 or 16 without index form scores 0; the question asked for index form.
- 6** (a) **72** **[1]** [2]
 (b) $0.6 \times 0.01 = 0.006$, then $0.006 \div 0.1 = 0.06$ **[1]**
0.0006 in (b) means they divided by 10 instead of by 0.1. This is the most common place-value error in the unit: dividing by a number smaller than 1 makes the result bigger.
- 7** (a) **0.050** **[1]** [2]
 (b) **0.0498** **[1]**
0.05 in (a) scores 0 — 3 d.p. requires the trailing zero. In (b), counting the leading zeros as significant gives 0.05, which is the error to look for.
- 8** all to 3 d.p.: 0.505, 0.550, 0.050, 0.500, 0.055 **[1]** [2]
0.05, 0.055, 0.5, 0.505, 0.55 **[1]**
Ordering by how long the number looks rather than by value gives 0.5, 0.05, 0.55, 0.055, 0.505. Padding to a common number of decimal places is the fix — insist on seeing it written down.

- 9** $7 \div 12 = 0.58333\dots$ **[1]** [2]
= 0.583 with a dot written above the final 3 only **[1]**
A dot over the 8, or over both 8 and 3, is the error — only the 3 repeats. Terminating the decimal at 0.58 also scores 1 at most.
- 10** (a) **1.17** **[1]** [2]
 (b) **0.935** **[1]**
1.065 in (b) is the sign error; 0.94 is a rounding habit that will wreck later multi-step work. Neither scores.
- 11** $\times 100$ gives 75 : 180 : 225 **[1]** [2]
 $\div$ HCF 15 gives **5 : 12 : 15** **[1]**
Stopping at 15 : 36 : 45 (divided by 5 only) scores 1. A ratio is simplest only when the three numbers share no factor at all.
- 12** (a) **T = c + mk** **[1]** (accept $T = mk + c$) [3]
 (b) $235 - 43 = 192$ **[1]**
 $192 \div 12 = 16$, so $m = ₹16$ per km **[1]**
Dividing 235 by 12 straight away gives 19.58 — a plausible-looking number, which is why it was chosen. The fixed fee must come off first.
- 13** (a) $15y - 10 - 8y + 14$ **[1] = 7y + 4** **[1]** [3]
 (b) **$x^2 - 6x$** **[1]**
 $7y - 24$ is what you get when -2×-7 is taken as -14 . Award the first mark only. This single sign is worth a whiteboard minute.
- 14** (a) **$6(2a + 3)$** **[1]** [3]
 (b) partial factorisation, e.g. $5x(3xy - 5y^2)$ **[1]**
 fully: **$5xy(3x - 5y)$** **[1]**
 $2(6a + 9)$ and $5(3x^2y - 5xy^2)$ are factorised but not fully. (a) is all-or-nothing; (b) gives 1 for a partial. The test: is anything left inside that both terms still share?
- 15** $\times 4$ on both sides: $5x - 3 = 4(x + 2)$ **[1]** [3]
 $5x - 3 = 4x + 8$ **[1]**
 $x = 11$ **[1]** check: $(55 - 3) \div 4 = 13$ and $11 + 2 = 13$ ✓
Multiplying only the x term by 4, or writing $5x - 3 = x + 8$, are the two failures. A non-integer answer here is the signal that a whole side was not multiplied.
- 16** (a) a square number has every prime index **even**; 3^5 has an odd index, so N is not square **[1]** [3]
 (b) one more factor of 3 makes that index even **[1]**
 $k = 3$ **[1]** ($Nk = 2^4 \times 3^6 \times 5^2 = 540^2$)
The discriminating question of Section 2. "It is too big" or "it ends in 0" scores nothing — the reason has to be about the indices. $k = 9$ or $k = 15$ means the rule is half-remembered.
- 17** (a) $36 \times 45 = 1620$ **[1]**; two decimal places altogether, so **1.62** **[1]** [3]
 (b) $\times 100$ on both: $12.6 \div 3 = 4.2$ **[1]**
16.2 or 0.162 in (a) is a place-count error, not a multiplication error — award the method mark and mark the placement wrong. In (b), 0.042 means only the divisor was scaled up.
- 18** $31/6 - 11/4$ **[1]** [3]
 common denominator 12: $62/12 - 33/12 = 29/12$ **[1]**
= 25/12 **[1]**
 $37/12$ appears when the whole numbers are subtracted separately and the negative fraction is flipped to keep it positive. Award 1. The improper-fraction route avoids it entirely — push it.

- 19** (a) $6 \times 21/8 = 126/8$ **[1] = 153/4 [1]** [3]
 (b) $8 \times 5/2 = 20$ **[1]**
125/8 in (a) is multiplying the whole number only and copying the fraction across — scores 0. In (b) an answer of 3.2 means they multiplied by 2/5 instead of dividing.
- 20** common denominator 360 **[1]** [3]
 $5/8 = 225/360$, $7/12 = 210/360$, $11/18 = 220/360$, $3/5 = 216/360$ **[1]**
5/8, 11/18, 3/5, 7/12 [1]
11/18 (0.611) and 3/5 (0.600) are deliberately close — swapping only those two scores 2 and is the near-miss to look for. Ordering by numerator or by denominator are the two wholesale errors.
- 21** $A : B = 9 : 12$ and $B : C = 12 : 10$, so $A : B : C = 9 : 12 : 10$ **[1]** [3]
 31 parts; $9300 \div 31 = ₹300$ per part **[1]**
Amir ₹2700, Bina ₹3600, Cara ₹3000 [1] (sum 9300 ✓)
Jamming the ratios together as $3 : 4 : 5$ without scaling B gives $2325 : 3100 : 3875$ — which still adds to 9300, so it survives the obvious check. That is exactly why the question is built this way: the shared number must match in both ratios.
- 22** (a) $8400 \times 0.85 = 7140$ **[1]**; $7140 \times 1.15 = ₹8211$ **[1]** [3]
 (b) $0.85 \times 1.15 = 0.9775$, so a **2.25% decrease [1]**
“Back to ₹8400, no change” is the error this question exists to catch — the 15% rise acts on 7140, not on 8400. Do not accept “2.25%” without the word decrease.
- 23** (a) $1250 \div 25 = 50$ bottles per minute **[1]**; $50 \times 72 = 3600$ bottles **[1]** [5]
 (b) second machine: $50 \times 3/5 = 30$ per minute **[1]**
 together: $50 + 30 = 80$ per minute **[1]**
 $6400 \div 80 = 80$ minutes **[1]**
Two traps. Using 1 hour instead of 72 minutes in (a) gives 3000. In (b), averaging the rates to 40 instead of adding them to 80 gives 160 minutes — rates add, they never average.
- 24** (a) $2[(3x + 2) + (x - 1)] = 58$ **[1]** [5]
 $8x + 2 = 58$ **[1]**, so $x = 7$ **[1]**
 (b) length 23 cm, width 6 cm **[1]**
 area = $23 \times 6 = 138 \text{ cm}^2$ **[1]**
Adding the two expressions and forgetting to double gives $x = 14.25$. A non-integer x in a question with tidy whole-number lengths is itself a warning sign — teach that as the check. Missing cm^2 in (b) loses the last mark.
- 25** (a) $448 \div 1.12$ **[1] = 400 members [1]** [5]
 (b) juniors = $448 \times 5/8 = 280$, seniors = 168 **[1]**
 $280 \times 1.2 = 336$ and $168 \times 0.75 = 126$ **[1]**
 $336 : 126 = 8 : 3$ ($\div 42$) **[1]**
 $448 \times 0.88 = 394.24$ is the reverse-percentage error and scores 0 in (a): a 12% rise is undone by dividing by 1.12, not by taking 12% off. In (b), leaving $336 : 126$ unsimplified scores 2 of 3.
- 26** (a) $2.4 \div 0.08 = 240 \div 8$ **[1] = 30 sheets [1]** [5]
 (b) $30 \times 4.6 = 138 \text{ g}$ **[1]**
 $138 \div 1000 = 0.138 \text{ kg}$ **[1]**
= 0.14 kg (2 s.f.) [1]
0.3 or 3 sheets in (a) is a place-value slip in the division, and it is one the student can catch himself — a stack cannot hold a fraction of a sheet. Leaving 0.138 unrounded in (b) loses only the final mark.

- 27** (a) dividing by 0.1 **multiplies by 10**; he divided by 10 instead **[1]**; correct answer **48** [5]
[1]
(b) he divided by 2 only, but 6, 9 and 15 still share a factor of 3 **[1]**; correct answer **2 : 3 : 5** **[1]**
(c) **0.28** **[1]** ($7 \times 4 = 28$, and two decimal places in the question means two in the answer)

The point here is naming the rule, not producing the number. A correct answer with no explanation scores 1 of 2 in (a) and (b). If he can say “dividing by something less than one makes it bigger” in his own words, that is the whole place-value unit landing.